

Total No. of Questions—8]

[Total No. of Printed Pages—4

Seat No.	
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S.E. (Computer Engg.) (First Semester) EXAMINATION, 2014

OPERATING SYSTEM AND ADMINISTRATION

(2012 COURSE)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Attempt Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4,
Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

(iv) Assume suitable data, if necessary.

1. (a) Define the following OS terms : [6]

(i) Multiprogramming

(ii) Multitasking

(iii) Multiprocessing

(iv) Multithreading

(v) Monolithic kernel

(vi) Microkernel.

(b) Explain the file system data structures of Unix with neat diagram
(UFDT, FT and IT). [6]

P.T.O.

Or

2. (a) Draw and explain the block diagram of system kernel. [6]
- (b) Draw and explain the 4 state process diagram (inclusive of kernel and user modes). [6]
3. (a) Explain the different types of files in unix. Also give the command to display the same. List the different categories of users in unix. [6]
- (b) Explain the following commands in unix :
grep, ripe, stat, mount, chmod and ln. [6]

Or

4. (a) Differentiate between Grub and Lilo. [3]
- (b) Explain the run levels of Booting process in unix. [6]
- (c) Explain any *three* characteristics of perl programming. [3]
5. (a) Explain in detail signal and kill function with parameters. [4]
- (b) Explain the following unix commands :
nice, renice, top, prstat, strace. [5]

(c) Differentiate between the following : [4]

(i) Modern Access Control and Real World Access Control

(ii) User mode and Kernel mode.

Or

6. (a) Explain the concept of : [6]

(i) nobody account

(ii) real, effective and save id's

(iii) setuid and setgid commands.

(b) Explain runaway processes, sudo command and su command. [3]

(c) Explain the concept of : [4]

(i) shadow password

(ii) pseudo users with respect to /bin/false and /bin/nologin.

7. (a) Give the entries of /etc/passwd file and /etc/group file. [6]

(b) Explain the following : [6]

(i) fsck

(ii) mkfs

(iii) file system mounting.

(c) Define disc partitioning. [1]

Or

8. (a) Explain the following : [6]
- (i) useradd
 - (ii) userdel
 - (iii) RAID.
- (b) List any *three* storage devices. [3]
- (c) Differentiate between the following : [4]
- (i) SATA and PATA
 - (ii) Unix and Linux.